

1 Problem 1

Prove that convergence of $\{s_n\}$ implies convergence of $\{|s_n|\}$. Is the converse true? (Baby Rudin, Problem 3.1, p.78)

2 Problem 2

Calculate $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$. (Baby Rudin, Problem 3.2, p.78)

3 Problem 3

If $s_1 = \sqrt{2}$, and $s_{n+1} = \sqrt{s + \sqrt{s}}$ ($n = 1, 2, 3, \dots$), prove that s_n converges, and that $s_n < 2$ for $n = 1, 2, 3, \dots$ (Baby Rudin, Problem 3.3, p.78)

4 Problem 4

Find the upper and lower limits of the sequence $\{s_n\}$ defined by $s_1 = 0; s_{2m} = \frac{s_{2m-1}}{2}; s_{2m+1} = \frac{1}{2} + s_{2m}$. (Baby Rudin, Problem 3.4, p.78)

5 Problem 5

For any two real sequences $\{a_n\}$, $\{b_n\}$, prove that

$$\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n, \quad (1)$$

provided the sum on the right is not of the form $\infty - \infty$. (Baby Rudin, Problem 3.5, p.78)

6 Problem 6

Investigate the behavior (convergence or divergence) of $\sum a_n$ if

(a) $a_n = \sqrt{n+1} - \sqrt{n}$

(b) $a_n = \frac{\sqrt{n+1} - \sqrt{n}}{n}$

(c) $(\sqrt[n]{n} - 1)^n$

(d) $a_n = \frac{1}{1+z^n}$ for complex z .

(Baby Rudin, Problem 3.6, p.78)